INCH-POUND

MIL-PRF-55339C
w/ AMENDMENT 4
19 May 2023
SUPERSEDING
MIL-PRF-55339C
w/ AMENDMENT 3
08 December 2018

PERFORMANCE SPECIFICATION

ADAPTERS, CONNECTORS, COAXIAL, RADIO FREQUENCY, (BETWEEN SERIES AND WITHIN SERIES), GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements and tests for between series and within series, radio frequency (RF), coaxial connector adapters.

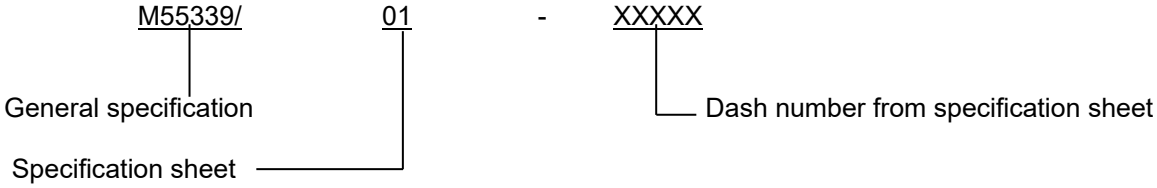
1.2 Classification. Adapters consist of the following classes with the Part Identifying Number (PIN), as specified (see 3.1).

a. Class 1 – A class 1 adapter is an adapter which is intended to provide superior RF performance at specified frequencies, and for which all RF characteristics are completely defined.

b. Class 2 – A class 2 adapter is intended to provide mechanical connection within an RF circuit providing specified RF performance.

1.2.1 PIN. The PIN consists of the letter “M” followed by the basic specification sheet number. The first digit in the dash number designates the material and finish of the adapter body (shell); i.e., “0” for silver plated brass, “3” for passivated corrosion-resistant steel, “4” for gold plated copper beryllium, “6” to be used between series SMA and other series (SMA body to be corrosion resistant steel and the other series to be brass), or "7" for nickel plated brass. Succeeding digits will be assigned to designate the former ‘UG’ number or a nonsignificant number, as applicable.

Example:



Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Columbus, ATTN: VAI, P.O. Box 3990, Columbus OH 43218-3990 or email to RFConnector@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59588 - Rubber, Silicone.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-39012 - Connectors, Coaxial, Radio Frequency, General Specification For

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1285 - Marking of Electrical and Electronic Parts
MIL-STD-202 - Electronic and Electrical Component Parts.
MIL-STD-202-101 - Salt Atmosphere (Corrosion)
MIL-STD-202-106 - Moisture Resistance
MIL-STD-202-107 - Thermal Shock
MIL-STD-202-112 - Seal
MIL-STD-202-201 - Vibration
MIL-STD-202-213 - Shock (Specified Pulse)
MIL-STD-202-301 - Dielectric Withstanding Voltage
MIL-STD-202-302 - Insulation Resistance
MIL-STD-348 - Radio Frequency Connector Interfaces for MIL-DTL-3643, MIL-DTL-3650, MIL-DTL-3655, MIL-DTL-25516, MIL-PRF-31031, MIL-PRF-39012, MIL-PRF-49142, MIL-PRF-55339, MIL-DTL-83517.
MIL-STD-790 - Established Reliability and High Reliability Qualified Products List (QPL) Systems for Electrical, Electronic, and Fiber Optic Parts Specifications.
MIL-STD-889 - Galvanic Compatibility of Electrically Conductive Materials

(Copies of these documents are available online at https://quicksearch.dla.mil/ or https://assist.dla.mil.)

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2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEFENSE STANDARDIZATION PROGRAM OFFICE

SD-6 - Provisions Governing Qualification

(Copies of these documents are available online at https://assist.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.
ASTM INTERNATIONAL

ASTM A342/A342M - Standard Test Methods for Permeability of Feebly Magnetic Materials
ASTM A484/B484M - Steel, Bars, Billets and Forgings, Stainless
ASTM A582/B582M - Free-Machining Stainless and Heat-Resisting Steel Bars
ASTM B16/B16M - Rod, Brass, Free-Cutting, Bar and Shapes for use in Screw Machines
ASTM B36/B36M - Plate, Brass, Sheet, Strip, and Rolled Bar
ASTM B88 - Tube, Water, Seamless Copper
ASTM B121/B121M - Plate, Leaded Brass, Sheet, Strip, and Rolled Bar
ASTM B124/B124M - Copper and Copper Alloy Forging Rod, Bar, and Shapes
ASTM B139/B139M - Rod, Phosphor Bronze, Bar, and Shapes
ASTM B152/B152M - Copper Sheet, Strip, Plate, and Rolled Bar
ASTM B194 - Copper-Beryllium Alloy Plate, Sheet, Strip and Rolled Bar
ASTM B196/B196M - Rod and Bar, Copper-Beryllium Alloy
ASTM B197/B197M - Wire, Alloy, Copper-Beryllium
ASTM B455 - Standard Specification for Copper-Zinc-Lead Alloy Extruded Shapes.
ASTM B488 - Gold for Engineering Uses, Electrodeposited Coatings of
ASTM B700 - Electrodeposited Coatings of Silver for Engineering Uses
ASTM D2116 - Molding and Extrusion Materials, FEP-Fluorocarbon
ASTM D4894 - Polytetrafluoroethylene (PTFE) Granular Molding and RAM Extrusion Materials
ASTM D4895 - Polytetrafluoroethylene (PTFE) Resins Produced From Dispersion

(Copies of these documents are available from https://www.astm.org.)

ASME INTERNATIONAL

ASME B46.1 – Surface Texture (Surface Roughness, Waviness, and Lay.

(Copies of these documents are available online at https://www.asme.org.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS, INC. (IEEE)

IEEE 287 – Standard For Precision Coaxial Connectors.

(Copies of these documents are available online from https://www.corporate-communication@ieee.org.)

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INTERNATIONAL ORGANIZATION FOR STANDARDS (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories.

(Copies of this document is available online at https://www.iso.org/iso/home.html)

SAE INTERNATIONAL

SAE-AMS-QQ-N-290	- Nickel Plating (Electrodepositied).
SAE-AMS2700	- Steels, Passivation of Corrosion Resistant.
SAE-EIA557	- Statistical Process Control Systems.

(Copies of these documents are available online at https://www.sae.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Adapters furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3).

3.2.1 Qualified Products List (QPL) system. The manufacturer shall establish and maintain a system to meet the requirements of MIL-STD-790 and the requirements herein (see 3.1). Evidence of such compliance shall be verified by the qualifying activity as a prerequisite for qualification and retention of qualification. In addition, the manufacturer shall establish a Statistical Process Control (SPC) system that meets the requirements specified in SAE-EIA557.

3.2.2 Design/process change. After qualification, if the manufacturer seeks to A) makes design changes or process changes to qualified product, B) use alternate / substitute materials, or C) use alternate suppliers/vendors, then the manufacturer shall propose a design/construction change plan. The plan will be coordinated with the qualifying activity to construct a requalification test plan. Approval for proposed changes must be approved by the qualifying activity prior to implementing the design or process change.

3.3 Critical interface material. Material shall be specified herein and in TABLE I. If materials other than those specified are used, the contractor shall certify to the qualifying activity that the substitute material enables the adapters to meet the requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the product. When a definite material is not specified, a material shall be used which will enable the adapters to meet the requirements of this specification. Any substitute materials shall be non-hazardous and permit the adapter to meet the requirements of section 3 of this specification.

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TABLE I. Materials.

Component material	Applicable specification
Steel	ASTM A484/B484M, ASTM A582/B582M
Brass	ASTM B16/B16M, ASTM B36/B36M, ASTM B121/B121M
Copper beryllium	ASTM B194, ASTM B196/B196M, ASTM B197/B197M, ASTM B455
Phosphor bronze	ASTM B139/B139M
Soft copper	ASTM B152/B152M
Copper	ASTM B88, ASTM B124/B124M
PTFE fluorocarbon	ASTM D4894 and ASTM D4895
FEP fluorocarbon	ASTM D2116
Silicon rubber	A-A-59588

3.3.1 Critical interface materials and finish. Unless otherwise specified (see 3.1), adapter center contacts and bodies shall be plated in the following manner in order to meet the requirements of this specification and avoid detrimental interactions between dissimilar metals.

3.3.1.1 Center contacts. The male pin shall be plated to a minimum gold thickness of 50 micro inches (1.27 micro meters) in accordance with ASTM B488, type II code C, class 1.27, over 50 micro inches (1.27 micro meters) minimum of nickel in accordance with SAE-AMS-QQ-N-290, class 1, measured anywhere along the mating surface, for all series. The socket contact shall be plated to a minimum of 50 micro inches (1.27 micro meters) of gold in accordance with ASTM B488, type II, code C, class 1.27, over 50 micro inches minimum of nickel in accordance with SAE-AMS-QQ-N-290, class 1, including the I.D., measured at a depth of .040 inch (1.01mm) minimum. The plating on non-significant surfaces in the I.D. shall be of sufficient thickness to ensure plating continuity and uniform utility and protection. This plating may consist of an underplate only. A silver underplate shall not be permitted on any contact, pin or socket.

3.3.1.2 Adapter bodies. All brass bodied adapters shall be silver plated in accordance with ASTM B700 to a minimum thickness of 0.000200 inch (0.005 mm) over a copper underplate, or nickel plated in accordance with SAE-AMS-QQ-N-290, 0.000200 inch (0.005mm), minimum. All copper beryllium bodied adapters shall be gold plated to a minimum thickness of 50 micro inches (1.27 μm) in accordance with ASTM B488, type II, code C, class 1.27 over a copper flash. All corrosion resistant steel bodied adapters shall be passivated in accordance with SAE-AMS2700, Method 2, unless otherwise specified (see 3.1). NOTE: Ferrous or nickel alloys shall not be used on brass or copper beryllium bodied adapters (i.e., coupling nut, etc.).

3.3.2 Dissimilar metals. Dissimilar metals between which an electromotive couple may exist shall not be placed in contact with each other. Reference is made to MIL-STD-889 for definition of dissimilar metals.

3.3.3 Nonmagnetic materials. All parts (except hermetic sealed adapters and connector hardware) shall be made from materials which are classed nonmagnetic (see 3.9).

3.3.4 Spring members. Unless otherwise specified (see 3.1), center contact spring members shall be made of copper beryllium.

3.3.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.4 Configuration and features. Adapters shall be of the configuration and physical dimensions specified (see 3.1). The nominal impedance, working voltage, frequency range, and temperature range shall be as specified (see 3.1).

3.4.1 Mating (visual indication). When applicable, a visual means for indicating complete mating shall be as specified (see 3.1).

3.4.2 Screw threads. Screw threads shall be in accordance with FED-STD-H28 unless otherwise specified (see 3.1).

3.4.3 Adapter interfaces. Adapter interfaces shall be in accordance with MIL-STD-348 unless otherwise specified (see 3.1).

3.5 Center contact retention. When adapters are tested as specified in 4.5.2, the center contacts shall withstand the axial force specified (see 3.1). The center contact shall meet the mating interface dimensions. When a torque requirement is specified (see 3.1) there shall be no rotation of the center contact.

3.6 Force to engage/disengage (see 4.5.3).

3.6.1 Bayonet and threaded types. When tested as specified in 4.5.3.1, the torque necessary to completely couple or uncouple the adapters shall not exceed that specified (see 3.1). Also the longitudinal force necessary to initiate the engaging or disengaging cycle shall not exceed that specified (see 3.1).

3.6.2 “Push on” adapter types (with or without detents). When tested as specified in 4.5.3.2, the forces necessary to fully engage or disengage the adapter shall not exceed that specified (see 3.1).

3.7 Coupling proof torque (threaded adapters). When tested as specified in 4.5.4, the coupling mechanism shall not be dislodged, and the adapter shall meet requirements of 3.6.1. The interface dimensions of the adapter shall remain as specified (see MIL-STD-348).

3.8 Mating characteristics. When adapters are tested as specified in 4.5.5, the mating dimensions shall be gauged as specified (see 3.1) and the dimensions shall remain within the specified tolerances (see 3.1).

3.9 Permeability of nonmagnetic materials. When adapters (except hermetic sealed and connector hardware) are tested as specified in 4.5.6 , the permeability (Mu) shall be less than 2.0.

3.10 Hermetic seal. When adapters are tested as specified in 4.5.7, the leakage rate shall not exceed that specified (see 3.1).

3.10.1 Leakage. When adapters are tested as specified in 4.5.7.1, there shall be no leakage as detected by escaping air bubbles.

3.11 Insulation resistance. When adapters are tested as specified in 4.5.8 , the insulation resistance shall not be less than that specified (see 3.1).

3.12 Voltage standing wave ratio (VSWR). When adapters are tested as specified in 4.5.9, the VSWR shall not exceed that specified over the frequency range specified (see 3.1).

3.13 RF leakage. When adapters are tested as specified in 4.5.10, the total leakage shall not exceed that specified (see 3.1).

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3.14 RF insertion loss. When adapters are tested as specified in 4.5.11, the insertion loss shall not exceed that specified (see 3.1).

3.15 Adapter durability. When adapters are tested as specified in 4.5.12, they shall show no evidence of severe mechanical damage and the coupling device shall remain functional. Adapters shall meet the applicable requirements of 3.6 and 3.8.

3.16 Dielectric withstanding voltage. When adapters are tested as specified in 4.5.13, there shall be no evidence of breakdown.

3.17 Contact resistance. When adapters are tested as specified in 4.5.14, the contact resistance of the center contact and outer contact shall be as specified (see 3.1).

3.18 Vibration. When adapters are tested as specified in 4.5.15, there shall be no electrical interruptions exceeding 1 microsecond (μs), or as otherwise specified (see 3.1). There shall be no evidence of visual mechanical damage after the test, and the contact resistance of the center contact shall not exceed the specified amount (see 3.1 and 3.17).

3.19 Shock (specified pulse). When adapters are tested as specified in 4.5.16, there shall be no electrical interruptions exceeding 1 μs unless otherwise specified (see 3.1). There shall be no evidence of visual or mechanical damage after the test, and the contact resistance of the center contact shall not exceed the specified amount (see 3.1).

3.20 Thermal shock. After testing as specified in 4.5.17, there shall be no evidence of visual mechanical damage to the adapter and it shall meet the dielectric withstanding voltage requirement (see 3.16), and the contact resistance specified for the center contact shall not be exceeded (see 3.1 and 3.17).

3.21 Moisture resistance. When adapters are tested as specified in 4.5.18 , there shall be no evidence of damage. They shall withstand the dielectric withstanding voltage specified (see 3.16), and the insulation resistance shall not be less than that specified (see 3.11).

3.22 Corona level. When adapters are tested as specified in 4.5.19, at the altitude and voltage specified (see 3.1), there shall be no evidence of sustained corona discharge.

3.23 RF high potential withstanding voltage. When adapters are tested as specified in 4.5.20, there shall be no breakdown, and the leakage current specified shall not be exceeded (see 3.1).

3.24 Corrosion (Salt spray). When adapters are tested as specified in 4.5.21, there shall be no exposure of the base metal on the interface or mating surface, and they shall meet the requirements of 3.6.1 or 3.6.2 as applicable.

3.25 Coupling mechanism retention force. When tested as specified in 4.5.22, the coupling mechanism shall not be dislodged from the adapter and shall be capable of meeting the requirements of 3.6.1 immediately after the test.

3.26 Marking. Adapters and associated fittings shall be permanently and legibly marked in accordance with the general marking requirements of MIL-STD-1285 with the PIN (see 1.2.1), manufacturer's federal supply code, and final assembly date code. The marking location is optional; when practicable, a location should be picked that will allow the marking to be visible after installation. Marking is required on all parts manufactured to this specification unless specifically excepted (see 3.1).

3.27 Workmanship. Adapters and associated fittings shall be processed in such a manner as to be uniform in quality and shall be free from sharp edges, burrs and other defects that will affect life, serviceability or appearance.

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3.28 Safety wire hole pullout. When applicable (see 3.1), adapters are to be tested as specified in 4.5.23. There shall be no evidence of hole tear out.

3.29 Change effectivity. Unless otherwise specified by the preparing activity and/or the qualifying activity, all changes from the previous revision of MIL-DTL-55339 shall become effective within 90 days from the date of publication of the latest revision. If unable to implement changes within the 90-day time-period, coordinate with the qualifying activity for additional guidance. Manufacturers that are QPL listed and have concerns regarding possible changes to retention reporting requirements should contact the qualifying activity for clarification.

3.30 Disposition of stock. Qualified manufacturers and their distributors may ship complete connectors and accessories until existing stock is depleted. The connectors and accessories shall meet all the requirements in accordance with the previous revision of this document. The preparing activity and/or the qualifying activity may specify changes as needed to ensure qualified material is received

3.31 Letters of interpretation and policy. Letters of interpretation and policy applicable to this document shall be approved in writing by the preparing activity. All letters of interpretation and policy applicable to MIL-DTL-55339 written prior to the current date of this document are not applicable to this revision. All subsequent letters of interpretation and policy letters are valid only until the next document change action (amendment or revision).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Conformance inspection (see 4.4).
c. Periodic inspection (see 4.4.2).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the “GENERAL REQUIREMENTS” of MIL-STD-202. For each test of threaded coupling connectors, where the test is performed on mated pairs, the pair shall be torqued to the specified value (see 3.1).

4.2.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with ISO/IEC 17025.

4.2.1.1 QPL system. The manufacturer shall establish and maintain a system to meet the requirements of MIL-STD-790 and the requirements herein (see 3.2.1). Evidence of such compliance shall be verified by the qualifying activity as a prerequisite for qualification and retention of qualification. Assembly plants shall be approved by the qualifying activity and listed on the applicable Qualified Products List. The qualified connector manufacturer shall certify that the assembly plant is approved for the distribution of the manufacturer’s parts. Assembly plants shall use only piece parts (i.e. subcomponents such as connector shells and inserts) produced by the qualified manufacturer. No incoming inspection other than visual examination is required of certified piece parts obtained from the qualified manufacturer, except when there is cause for rejection. Assemblies produced at the assembly plant shall be subjected to inspection to ensure that the assembly process conforms with that established at the qualified manufacturing plant. Quality control requirements, including Government inspection surveillance, shall be the same as required for the qualified connector manufacturer.

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4.3 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production. After receipt of the letter of authorization from the activity responsible for qualification, the applicant shall submit to the qualifying activity a copy of the test report (certified by the Government inspector indicating the extent to which the tests were witnessed), together with samples required by the qualifying activity. Qualification inspection shall be performed in accordance with the provisions of SD-6.

4.3.1 Sample size. Ten class 1 adapters of the same PIN, or six class 2 adapters of the same PIN, shall be subjected to the qualification inspection.

4.3.2 Group qualification. For group qualification of all series of adapters covered by this specification (see 3.1). The Government reserves the right to authorize performance of any or all qualification inspections of additional types in the group that are considered necessary for qualification within each group.

4.3.3 Inspection routine. The sample shall be subjected to the inspections specified in table II. All sample units shall be subjected to the inspection of group I. The sample units shall then be divided equally into two groups of 5 units (class 1) or three units each (class 2) and subjected to the inspection for their particular group and in the sequence given for that group.

4.3.4 Failures. One or more failures shall be cause for refusal to grant qualification approval.

4.3.5 Retention of qualification. To retain qualification the qualified manufacturer shall periodically forward reports and form DD1718 to the Qualifying Activity (see 6.3) as follows:

a. At 24 month intervals, unless otherwise specified by the Qualifying Activity, the qualified manufacturer shall forward a report with a summary of the results of group A and B testing (see 4.5). The results of tests for all reworked lots shall be identified and accounted for.

Note: Failure to submit the report within thirty days after the end of each reporting period may result in a stop ship notification or a loss of qualification for that product.

b. At 36 month intervals, unless otherwise specified by the Qualifying Activity, the qualified manufacturer shall submit requalification, group C test reports (see 4.5.2), including the number and mode of failures. The summary shall include results of all qualification verification inspection testing performed during that period. Reports shall include certification that the qualified source continues to produce qualified product under the same conditions as originally qualified (i.e., same processes, materials, design, construction, etc., including manufacturing locations), unless such changes have been authorized by the Qualifying Activity.

Note: The qualified source shall immediately notify the Qualifying Activity at any time during the reporting period when inspection data indicates failure of the qualified product to meet the requirements of the specification. If the test results indicates nonconformance with the specification requirements, and corrective action acceptable to the Qualifying Activity has not been taken, then action may be taken to remove failing product from the QPL. In addition, failure to submit the report within thirty days after the end of each reporting period may result in a stop ship notification or loss of qualification for that product. In the event that no production occurred during the reporting period, the qualified manufacturer shall submit a report certifying that the company still has the capability and facilities required to produce the qualified product. If there has been no production during two consecutive reporting periods, the Qualifying Activity, at their discretion, may require the manufacturer submit a representative product of each type, class, etc., to retention testing in accordance with qualification inspection requirements.

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The Qualifying Activity may exercise the right to request testing of QPL parts in which the manufacture has qualified to and is listed on the QPL for, but have not provided sample parts for retention of qualification for one or more retention periods.

4.3.5.1 Quality issues. If a company is found to have major quality issues, or is in violation of any of the requirements of the subject programs, the Qualifying Activity may place the company on probation and/or remove them from the applicable program for non-compliance. The Qualifying Activity will notify the company by letter of the quality issues and/ or program violations that have placed them on probation or are cause for their removal from the applicable program.

4.3.5.2 Removal from the applicable program. A company does not need to be placed on probation prior to removal from the applicable program if it is determined by the Qualifying Activity that the quality issues are systemic in nature and/or program violations are such that the QML/QPL/QTSL/QSLD/QSLM products from this facility pose a serious risk to the end user.

4.3.5.3 Requalification. When a company is removed from the qualification program and wishes to be reinstated; the company will be required to complete the entire initial qualification procedure.

4.3.6 Extension of qualification. Manufacturers who have products listed on QPL-39012 and produce adapters of the same series, may apply to the qualifying activity for extension of qualification to this specification, provided the interfacial coupling, materials and plating of the adapter and connectors are identical, and the adapter successfully meets the requirements of groups I and II in table II.

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TABLE II. Qualification inspection.

Inspection	Requirement paragraph	Test method paragraph
Group I		
Visual and dimensional inspection:		
Visual inspection	3.1, 3.3, 3.4 3.26 and 3.27	4.5.1.1
Dimensional inspection	3.1	4.5.1.2
Center contact retention	3.5	4.5.2
Force to engage/disengage	3.6	4.5.3
Coupling proof torque (threaded adapters)	3.7	4.5.4
Mating characteristics	3.8	4.5.5
Permeability (not applicable to hermetic-sealed adapters)	3.9	4.5.6
Workmanship	3.27	4.5.1.1
Hermetic seal (pressurized adapters only)	3.10	4.5.7
Leakage (pressurized adapters only)	3.10.1	4.5.7.1
Insulation resistance	3.11	4.5.8
Group II		
VSWR	3.12	4.5.9
RF leakage 1/	3.13	4.5.10
RF insertion loss 1/	3.14	4.5.11
Adapter durability	3.15	4.5.12
Force to engage/disengage	3.6	4.5.3
Coupling proof torque (threaded adapters)	3.7	4.5.4
Mating characteristics	3.8	4.5.5
VSWR	3.12	4.5.9
Dielectric withstanding voltage	3.16	4.5.13
Group III		
Contact resistance (center contact)	3.17	4.5.14
Vibration, high frequency 1/	3.18	4.5.15
Contact resistance (center contact)	3.17	4.5.14
Shock (specified pulse) 1/	3.19	4.5.16
Contact resistance (center contact)	3.17	4.5.14
Dielectric withstanding voltage	3.16	4.5.13
Contact resistance (center contact)	3.17	4.5.14
Thermal shock (hermetic sealed adapters)	3.20	4.5.17
Dielectric withstanding voltage	3.16	4.5.13
Contact resistance (center contact)	3.17	4.5.14
Moisture resistance	3.21	4.5.18
Dielectric withstanding voltage	3.16	4.5.13
Corona level 1/	3.22	4.5.19
Hermetic seal (pressurized adapters only)	3.10	4.5.7
Leakage (pressurized adapters only)	3.10.1	4.5.7.1
RF high potential withstanding voltage 1/	3.23	4.5.20
Corrosion (Salt spray)	3.24	4.5.21
Force to engage/disengage	3.6	4.5.3
Coupling mechanism retention force (when applicable see 3.1)	3.25	4.5.22
Force to engage/disengage	3.6	4.5.3

1/ These tests to be performed only during initial qualification as long as the qualifying design and manufacturing process has not been changed.

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4.4 Conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.4.1.1 Inspection lot. An inspection lot shall consist of all adapters of the same PIN produced under essentially the same conditions, and offered for inspection at one time.

4.4.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table III in the order shown.

4.4.1.2.1 Sampling plan (group A). Table III, tests shall be performed on a production lot basis. Samples shall be selected in accordance with table IV. If one or more defects are found, the lot shall be screened for that particular defect and defects removed. A new sample of parts shall be selected in accordance with table IV and all group A tests again performed. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification. When defects have no effect on the use or performance of the subject parts, reworking of the parts may be permitted upon approval of both the Preparing and Qualifying activities. Such allowances are infrequent and only apply to particular circumstances

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure
Visual inspection	3.1, 3.3, 3.4 3.26 and 3.27	4.5.1.1	
Dimensional inspection	3.1	4.5.1.2	See table IV
Hermetic seal	3.10	4.5.7	
Leakage	3.10.1	4.5.7.1	
Dielectric withstanding voltage	3.16	4.5.13	

4.4.1.2.2 Visual inspection (group A inspection). Each adapter shall be visually examined for completeness, workmanship, and identification requirements. Attention shall be given to those assemblies that require a gasket to determine the condition of the gasket. Gaskets missing, twisted, buckled, kinked, or damaged in any way shall be cause for rejection.

TABLE IV. Group A Sampling Plan.

Lot size	Visual and mechanical inspection
1 to 19	All
20 to 280	20
281 to 1,200	47
1,201 to 3,200	53
3,201 to 10,000	68
10,001 to 35,000	77
35,001 to 150,000	96
150,001 to 500,000	119
500,001 and over	143

4.4.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table V in the order shown, and shall be made on sample units which have been subjected to and passed the group A inspection. Adapters having identical piece parts may be combined for lot purposes and shall be in proportion to the quantity of each PIN numbered adapter produced.

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4.4.1.3.1 Group B sampling plan. A sample of parts shall be randomly selected in accordance with table VI. If one or more defects are found, the lot shall be screened for that particular defect and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected and subjected to all tests in accordance with table V. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

TABLE V. Group B Inspection.

Inspection	Requirement paragraph	Test method paragraph
Center contact retention	3.5	4.5.2
Force to engage/disengage	3.6	4.5.3
Coupling proof torque (threaded adapters)	3.7	4.5.4
Force to engage/disengage (threaded adapters)	3.6.1	4.5.3.1
Mating characteristics	3.8	4.5.5
Permeability (not applicable to hermetic-sealed adapters or connector hardware)	3.9	4.5.6
Insulation resistance	3.11	4.5.8
VSWR	3.12	4.5.9
Contact resistance (center and outer contacts)	3.17	4.5.14

TABLE VI. Group B Sampling Plan.

Lot size	Sample size	VSWR sample size
1 to 4	All	1
5 to 15	5	1
16 to 90	5	3
91 to 150	11	3
151 to 280	13	3
281 to 500	16	3
501 to 1,200	19	5
1,201 to 3,200	23	5
3,201 to 10,000	29	5
10,001 to 35,000	35	5
35,001 and over	40	8

4.4.1.3.2 Lot definition (group B inspection). The production lot, as defined in MIL-STD-790, is further specified as follows:

a. A production lot consists of all connectors covered under one military specification or standard, manufactured from the specified raw materials, processed under the same specification and procedures, produced by the same type of equipment, and submitted for inspection at one time. Each production lot of assembled connectors shall be a group identified by a common manufacturing record through all significant assembly operations.

b. Common manufacturing records and traceability shall begin with machining and molding and shall include connector assembly.

4.4.1.3.3 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract or purchase order, if the lot is accepted. Any adapter deformed or otherwise damaged during testing shall not be delivered on the contract or order.

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4.4.2 Periodic inspection. Periodic inspection shall consist of group C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.4.2.1.4), delivery of products which have passed groups A and B shall not be delayed pending the results of these periodic inspections.

4.4.2.1 Group C inspection. Group C inspection shall consist of the inspections specified in table VII, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspection.

4.4.2.1.1 Sampling plan. Six sample units of the same PIN shall be selected from the first lot produced after the date of notification of qualification. Thereafter, six sample units of the same part number shall be selected from current production after 200,000 adapters have been produced, or not less than once every three years, whichever occurs first. The sample units shall be divided equally and subjected to the inspections of the two subgroups.

4.4.2.1.2 Failures. If one or more sample units fail to pass group C inspection, the lot shall be considered to have failed.

4.4.2.1.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or order.

4.4.2.1.4 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subjected to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

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TABLE VII. Group C inspection. 1/

Inspection	Requirement Paragraph	Test method paragraph
Subgroup 1		
VSWR	3.12	4.5.9
RF leakage 2/	3.13	4.5.10
RF insertion loss 2/	3.14	4.5.11
Durability	3.15	4.5.12
Force to engage/disengage	3.6	4.5.3
Coupling proof torque	3.7	4.5.4
Mating characteristics	3.8	4.5.5
VSWR	3.12	4.5.9
Dielectric withstanding voltage	3.16	4.5.13
Subgroup 2		
Contact resistance (center and outer contacts)	3.17	4.5.14
Vibration, high frequency 2/	3.18	4.5.15
Contact resistance (center contact)	3.17	4.5.14
Shock (specified pulse) 2/	3.19	4.5.16
Contact resistance (center contact)	3.17	4.5.14
Dielectric withstanding voltage	3.16	4.5.13
Contact resistance (center contact)	3.17	4.5.14
Thermal shock (hermetic-sealed adapters)	3.20	4.5.17
Dielectric withstanding voltage	3.16	4.5.13
Contact resistance (center contact)	3.17	4.5.14
Moisture resistance	3.21	4.5.18
Dielectric withstanding voltage	3.16	4.5.13
Corona level 2/	3.22	4.5.19
Hermetic seal	3.10	4.5.7
Leakage	3.10.1	4.5.7.1
RF high potential withstanding voltage 2/	3.23	4.5.20
Corrosion (salt spray)	3.24	4.5.21
Force to engage/disengage	3.6	4.5.3
Coupling mechanism retention force	3.25	4.5.22
Force to engage/disengage	3.6	4.5.3

1/ Manufacturers who have products listed on QPL-39012 and produce adapters of the same series, may apply to the qualifying activity for a waiver in performing group C, subgroup 2, retention testing, providing the interfacial coupling, materials and plating of the adapter and connector are identical.

2/ These tests are to be performed during initial qualification only, as long as the qualifying design, materials and manufacturing process has not been changed.

4.5 Methods of inspection.

4.5.1 Test methods. The following identified tests and test methods assure adapter integrity within typical operating conditions and applications. Alternate commercial industry standard test methods are allowed; however when an alternate method is used, the qualifying activity must be notified prior to the performance of the test. The test methods described herein are proven methods and shall be the referee method in case of dispute.

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4.5.1.1 Visual and mechanical examination. Adapters and associated fittings shall be examined to verify that the design, construction, physical dimensions, assembly instructions, marking and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.26, and 3.27).

4.5.1.2 Dimensional examination. Mating dimensions shall be examined by mating the adapter with its applicable mating gauges or other suitable means acceptable to the Government.

NOTE: The documents listed in supplement 1 show the overall dimensions of the connector when assembled to the appropriate cable, and the detail mating dimensions at the interface. No other dimensions or details will be shown. The cable-end construction and other dimensions are controlled by performance requirements of the specification, i.e., cable retention, VSWR, RF leakage, etc., with cable in place.

4.5.2 Center contact retention (see 3.5). An axial force (see 3.1) shall be applied to the center contact of an unmated adapter. This force shall be applied at a rate of approximately 1 pound (4.448 N) per second until the specified force has been reached. The force shall be applied for a minimum period of 5 seconds. After removal of specified force, the axial location of the center contact at each end shall be determined. Each mating end of the adapter shall be tested. The torque specified (see 3.1) shall be applied to the center contact of an unmated adapter for a minimum period of 10 seconds. Each mated end of the adapter shall be tested.

4.5.3 Force to engage/disengage (see 3.6).

4.5.3.1 Bayonet and threaded types (see 3.6.1). The adapter shall be engaged with its mating standard part (see 3.1). During the entire coupling/uncoupling cycle (until the adapter is fully engaged/disengaged) the forces and/or torques necessary shall not exceed those specified (see 3.1). A thread coupled adapter is fully engaged with its mating standard part when their reference planes (see 3.1) coincide. A bayonet coupled adapter is fully engaged with its mating standard part when the bayonet studs have passed the detent and their reference planes coincide. No additional tightening torque shall be applied. The mating standard part is a steel jig containing the critical interface dimensions finished to the tolerances specified, or a qualified mating connector with the approval of the qualifying activity (see 3.1). Its spring members when applicable shall be heat treated beryllium copper. The surface finish or mating surfaces shall be 16 microinches maximum, in accordance with ASME B46.1.

4.5.3.2 “Push-on” connector types (see 3.6.2). The adapter under test shall be engaged with its standard mating part (gauge). During this engaging cycle the force necessary to fully engage the adapters shall not exceed that specified (see 3.1). Upon completion of engagement, an opposite force necessary for disengagement shall be applied. This force shall be within the limits specified, and shall include any unlatching forces required.

4.5.4 Coupling proof torque (see 3.7). The adapter under test shall be engaged with its mating standard part (gauge) and the coupling nut tightened to the torque value specified (see 3.1). After one minute the connector under test and its mating standard part shall be disengaged.

4.5.5 Mating characteristics (see 3.8). After insertion of the specified oversize pin the specified number of times (see 3.1), the contact to be tested shall be held rigid by means of a suitable jig or fixture, or a qualified mating connector with the approval of the qualifying activity. A gauge containing the test pin or test ring and a suitable force indicating dial shall be aligned to within 0.004 TIR of any plane passing through the axis of the contact under test. Engagement or withdrawal of the test pin or test ring shall be made smoothly and at such a rate that the dial does not bounce or otherwise give a false reading. The test pin or test ring may be chamfered to facilitate entry, but the specified engagement length shall not include the chamfer length and the finish shall be as specified and in accordance with ASME B46.1.

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4.5.6 Permeability of nonmagnetic materials (see 3.9). The permeability of the adapter shall be measured with an indicator conforming to ASTM A342/A342M.

4.5.7 Hermetic seal (see 3.10.). The unmated adapters shall be tested in accordance with, MIL-STD-202-112. The following details shall apply:

a. Method of mounting – in its normal manner in specified mounting hole (see 3.1).
b. Test condition letter. C.
c. Procedure number. I.
d. Leakage rate sensitivity: 10^{-8} cubic centimeters per second.

4.5.7.1 Leakage (pressurized) (see 3.10.1). The unmated adapter shall be mounted in its normal manner in the specified mounting hole (see 3.1) on a closed container. The specified air pressure (see 3.1) shall be applied to the interior of the container. The exposed portion of the adapter under pressure shall be fully immersed in water or alcohol-water mixture and observed for 1 minute minimum.

4.5.8 Insulation resistance (see 3.11). Adapters shall be tested in accordance with MIL-STD-202-302, test condition B. Measured between the center contact and body.

4.5.9 VSWR (see 3.12). The VSWR shall be measured in accordance with the following procedure or a method acceptable to the Government. In the event of dispute the method outlined herein shall be used. The measurement shall be made on a Vector Network Analyzer using a full 12 term error correction with the appropriate calibration standards. A minimum of 400 data points shall be used.

Optional calibration: single port 6 term calibration can be used.

The test adapter shall be terminated to an appropriate load. The load shall ensure that no signal is reflected back to the adapter under test.

4.5.10 RF leakage (see 3.13). The mating adapter pair to be tested shall be assembled as shown and tested on figure 1. The close fitting brass tubing shall be machined to thread into adapter in lieu of compression nut. This test setup between 500 MHz and 11 GHz, shall have a dynamic range from 20 dBm to better than 100 dBm or a difference of 90 dB. Using a + 20 dBm RF source with 10 dB isolation, an additional 30dB range can be obtained by use of attenuator pads or a step attenuator producing a total range of 120 dB. The shorting plunger is adjusted to produce a maximum reading in the detector with the triaxial assembly inserted. The insertion loss caused by the insertion of the triaxial assembly adjusted as shown is a measure of the total leakage of the mated adapter pair both at its interface.

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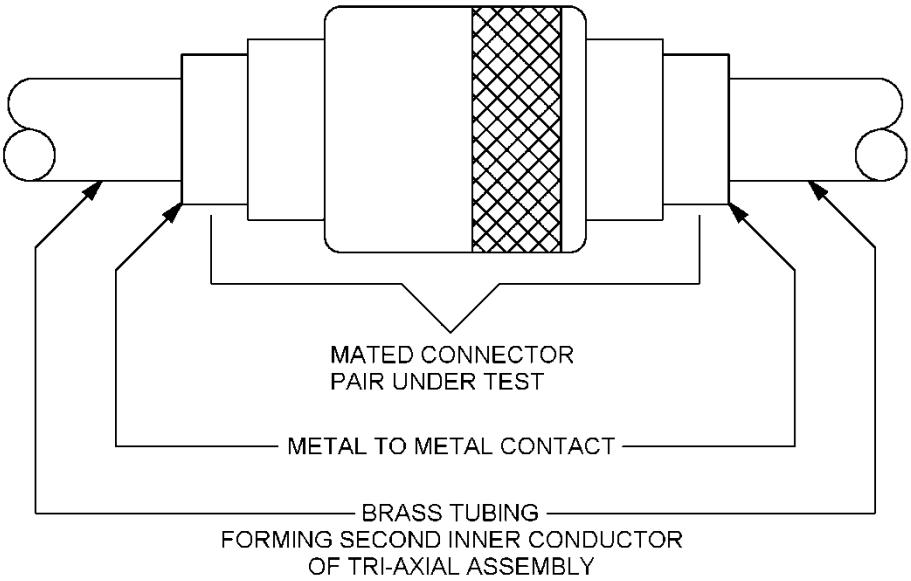


FIGURE 1. Connector assembly for RF leakage test.

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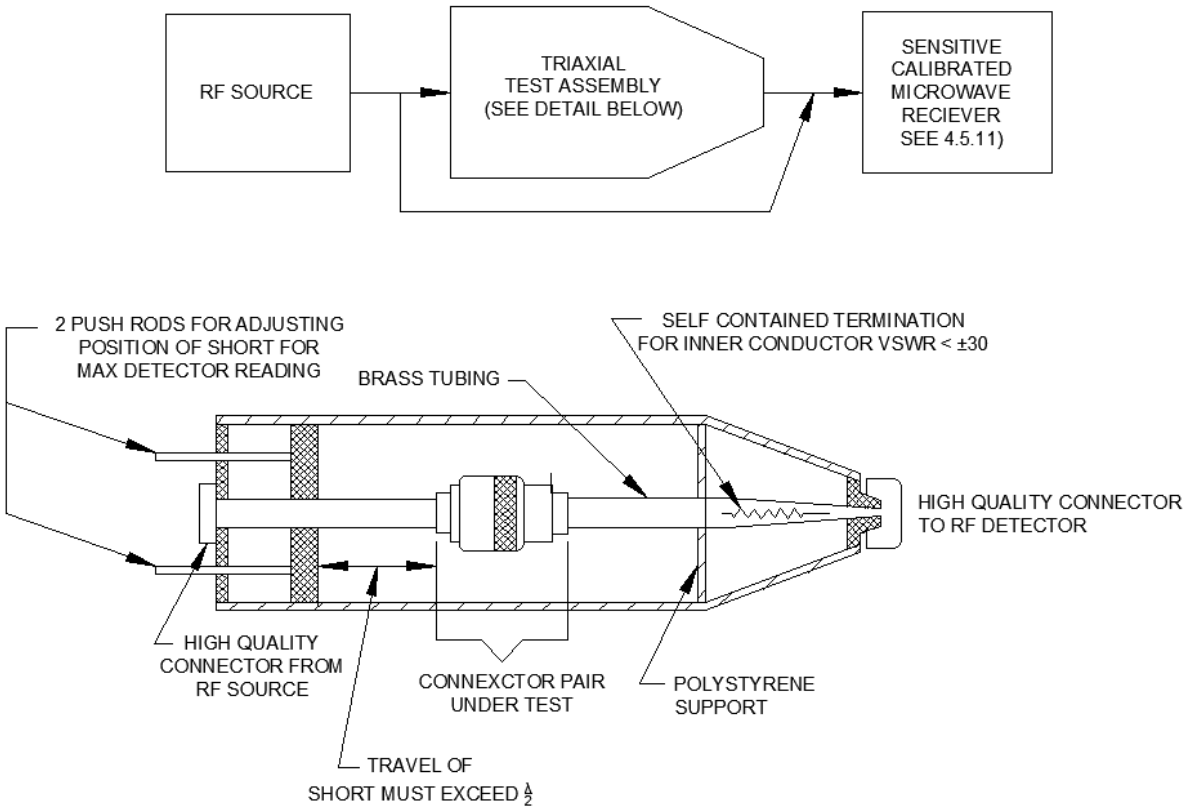


FIGURE 1. Connector assembly for RF leakage test - Continued.

4.5.11 RF insertion loss (see 3.14). The adapter shall be tested as shown on figure 2. Insertion loss of a mated adapter pair is defined as the increase of a loss due to insertion of a mated adapter pair in a cable; this includes the reflection losses to the cable and the dissipating losses in the pair. First insert cable assembly number 1 and tune out its input VSWR by means of tuner number 4 and balance setup. Then insert the cable assembly number 2 which includes the adapter pair under test. With tuner number 4 in the same position and the electrical length of the cable assembly the same as that of number 1, record increase of insertion loss; add to this as a correction the cable loss of the removed section due to the length $\Delta L1 + \Delta L2$ at this frequency. The sum of the increase and the correction is the insertion loss of the connector pair. Use of alternate test methods may be approved by the qualifying activity.

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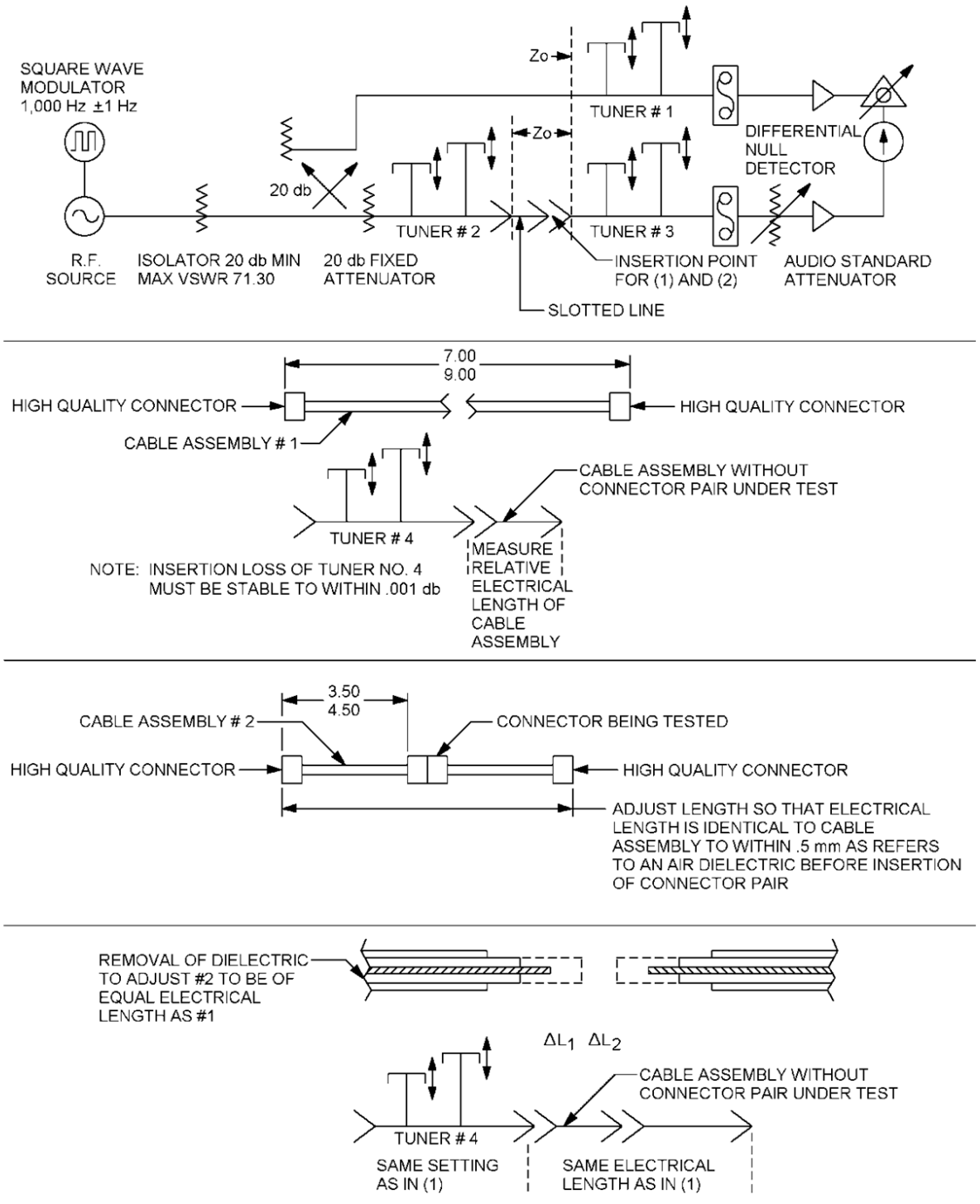


FIGURE 2. Method of insertion loss measurement of rated connector pair.

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4.5.12 Durability (see 3.15). Each adapter under test shall be mated with a typical production connector in accordance with this specification. The adapter shall be subjected to the number of cycles of mating and unmating specified (see 3.1). The adapter and its mating part shall be completely engaged and completely disengaged during this cycle. Lubrication of the threads or rotational parts shall not be employed for this test unless specified (see 3.1). It is permissible to shake or blow debris from the threads or interface surfaces at intervals of not less than 50 cycles. Solvents or tools shall not be used for cleaning.

4.5.13 Dielectric withstanding voltage (see 3.16). Adapters tested in accordance with MIL-STD-202-301. The following details shall apply:

a. Special preparations or conditions.

(1) The maximum relative humidity shall be 50 percent. When facilities are not available at this test condition, adapters are tested at room ambient relative humidity. In case of dispute, if the test has been made at room ambient relative humidity, retest shall be made at 50 percent maximum relative humidity.
(2) Precautions shall be taken to prevent air-gap voltage breakdowns.
(3) The voltage shall be metered on the high side of the transformer.

b. Magnitude of test voltage (see 3.1): The voltage shall be instantaneously applied.

c. Nature of potential: Alternating current.

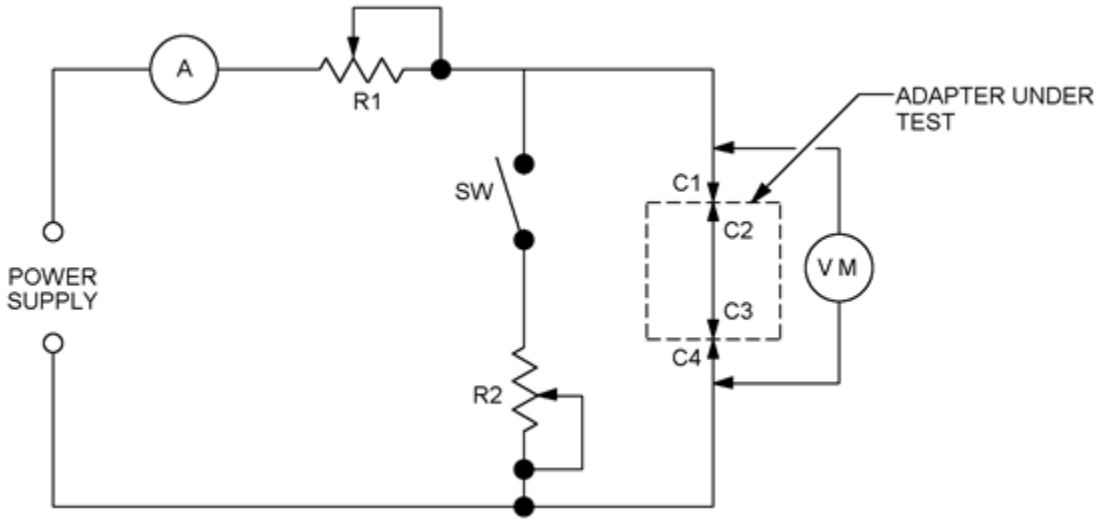
d. Points of application of test voltage: Between the center contact and body.

4.5.14 Contact resistance (see 3.17). All contact resistance tests shall be conducted with the apparatus shown on figure 3. Circuit adjustments and the measurement procedures for all contact resistance tests shall be in accordance with 4.5.14.1 The contact resistances to be measured are:

a. The contact resistance of the mated outer conductor contacts.
b. The contact resistance of the mated inner conductor contacts.
c. Measurement can be taken using a 4-Wire Digital Low Resistance Ohmmeter or equivalent.

4.5.14.1 General procedure. The apparatus shall be assembled as shown on figure 3. The contacts, C1 – C2, shown on the figure represent the mating contacts upon which millivolt drop tests are to be conducted.

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a. Remove contacts C2 – C3 from the measuring circuit.
b. Close switch (SW).
c. Adjust resistor (R2) for a voltmeter (VM) reading of 50 millivolts.
d. Connect contacts C2 – C3 to the measuring circuit.
e. Check to see that the voltage drops significantly prior to opening the switch in (f).
f. Open switch (SW).
g. Adjust resistor (R1) for a circuit current of 1 ampere.
h. Measure the voltage drop across contacts C1 – C4 and call this “e”.
i. Compute contact resistance as follows:
Contact resistance (milliohms) = e (in millivolts) ÷1 ampere.

A – Ammeter (0-1 ampere)
R1 – Variable resistor (0-15 ohms).
Sw – Switch (contact resistance <.01 ohm).
R2 – Variable resistor (0-0.1 ohm)
Vm – Voltmeter.
Power supply – DC 10 volts at 1 ampere.

FIGURE 3. Diagram for contact resistance.

4.5.15 Vibration, high frequency (see 3.18). The mated adapter shall be tested in accordance with MIL-STD-202-204. The following details and exceptions shall apply: Contacts may be connected in series.

a. Mating connector – Cabled connector in accordance with MIL-PRF-39012.
b. Mounting of specimens – The adapter shall be mounted by its normal mounting device and engaged by its normal coupling mechanism; safety wire shall not be used. Adapters having no provisions for mounting may be held to the jig (see figure 4) by a suitable clamp.
c. Electrical load – At least 100 milliamperes shall be flowing through each set of contacts.
d. Test condition – B.
e. Measurements – During vibration, continuity of the center and outer contacts shall be monitored with a detector capable of detecting interruptions of 1 microsecond or greater duration.

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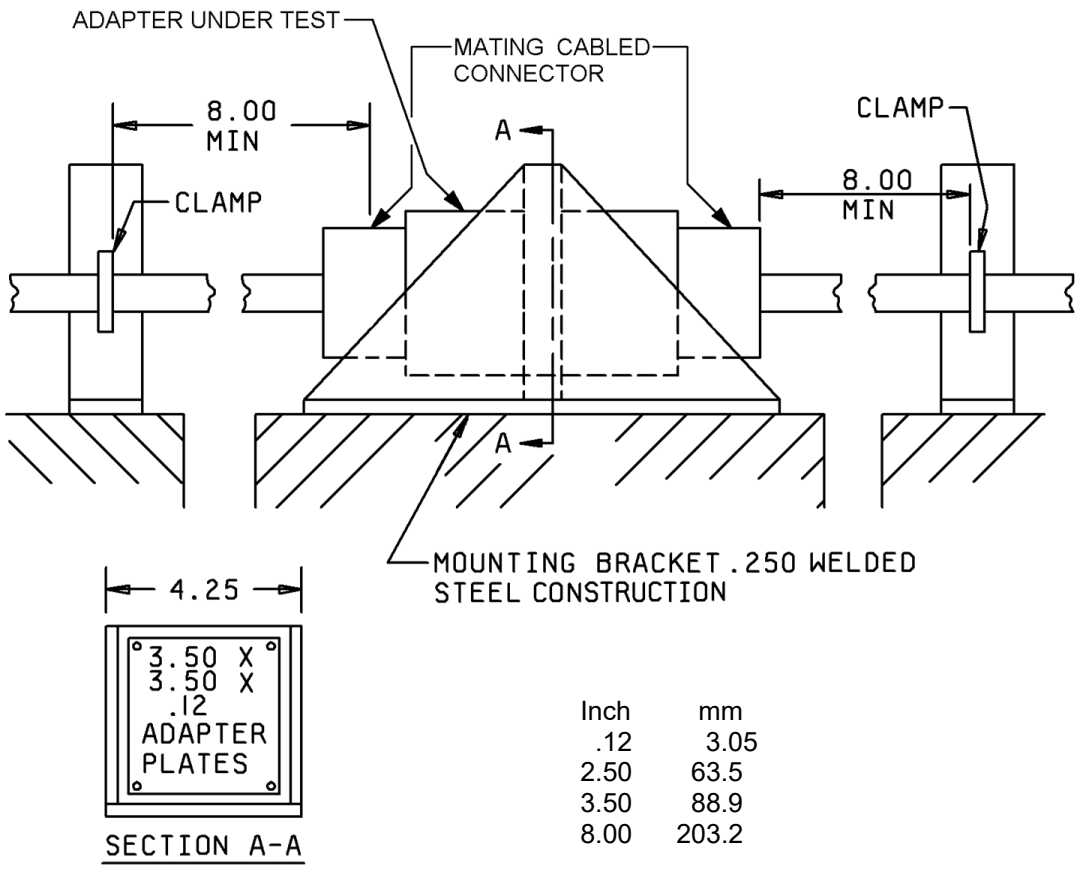


FIGURE 4. Vibration testing set-up.

4.5.16 Shock (specified pulse) (see 3.19). The adapter shall be tested in accordance with MIL-STD-202-213. The following exceptions and details shall apply:

a. Mating connector – Cabled connector per MIL-PRF-39012.

b. Mounting – Panel or bulkhead adapters shall be mounted by normal means. All other adapters shall be rigidly clamped to the shock table.

c. Test condition letter – As specified (see 3.1). Three shocks in each of three mutually perpendicular planes shall be applied; one of which shall be parallel to the axis of the adapter.

d. Measurements during shock – Continuity shall be monitored with a detector capable of detecting interruptions of 1 microsecond or greater in duration.

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4.5.17 Thermal shock (see 3.20). Adapters shall be tested in accordance with MIL-STD-202-107. The following details shall apply:

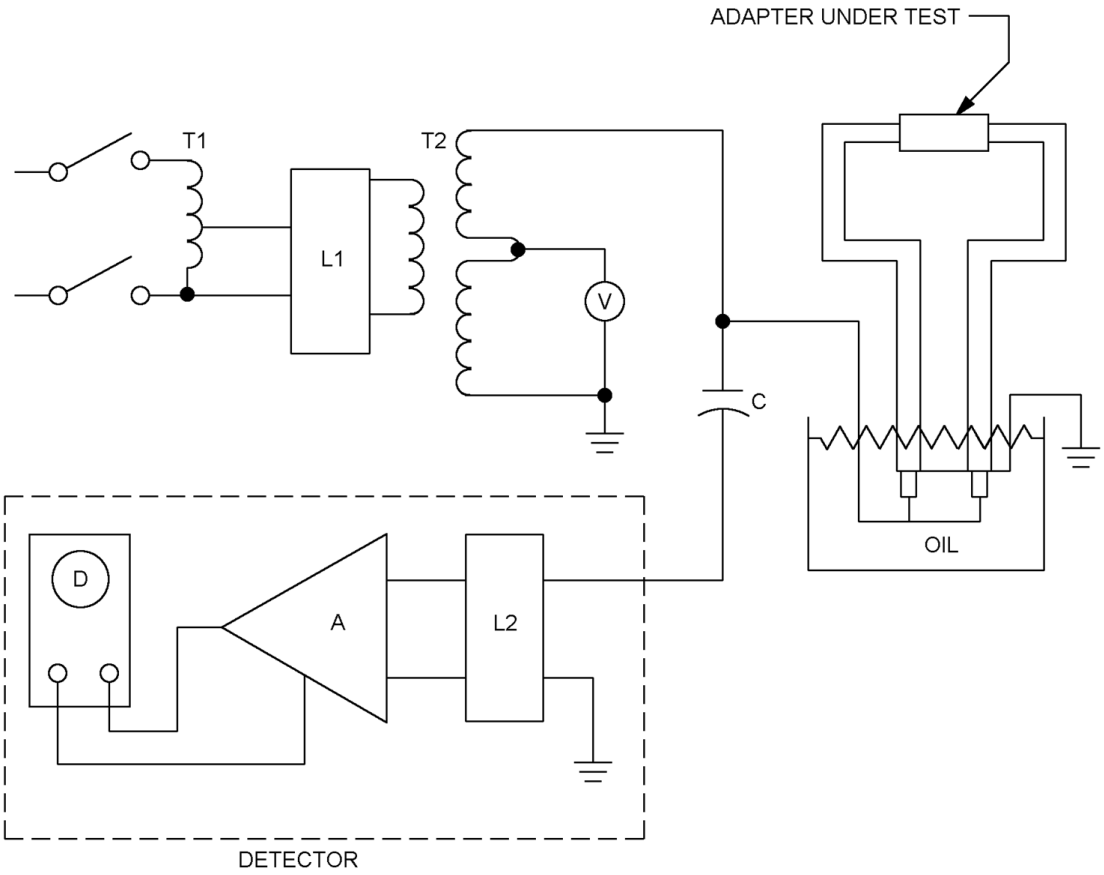
a. Test condition letter (see 3.1).
b. Measurements before and after cycling – Not applicable.

4.5.18 Moisture resistance (see 3.21). The adapter shall be tested in accordance with MIL-STD-202-106. The following exceptions and conditions shall apply:

a. Mating connector – Cabled connector in accordance with MIL-PRF-39012.
b. Initial measurements – Not applicable.
c. Loading voltage – Not applicable.
d. Number of cycles – 10 continuous cycles except step 7b (vibration) shall be omitted.
e. Final measurements – After the final cycle and within 5 minutes after removal from high humidity, insulation resistance shall be measured as specified in 4.5.8 .

4.5.19 Corona level (see 3.22). The test sample shall be connected to a mating connector and arranged in a suitable test circuit such as indicated on figure 5. Components of the test circuit shall be corona free to the extent that a discharge of five picocoulombs or less can be measured when the 60 Hz test potential is increased to the value specified at the reduced pressure specified (see 3.1). The type of cable and length of cable used shall be as specified (see 3.1). No grease or similar compounds shall be used in or on the test item. After the sample is purged of air, the 60 Hz voltage shall be slowly increased until the detector, operated at a sensitivity of five picocoulombs, indicates a sustained corona discharge. The voltage shall then be decreased until corona is at the five picocoulombs level or less. The latter value is being the corona level of the connector under test. The contractor may, at his own option, use a corona detector (which has been approved by the Government) for performing the test in lieu of the test set up on figure 5.

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C – Corona free coupling capacitor (see note 1).
D – Discharge display.
L1 – Input line filter (see note 2).
L2 – 10-50 kHz detector input filter.
A – Detector amplifier.
T1 – 0-130 V variable transformer.
T2 – High voltage transformer (corona free – less than 5 picocoulombs).
V – Voltmeter.

NOTES:
1. Equal to or greater than total circuit capacitance.
2. 100 dB 14 kHz to 10 GHz.

FIGURE 5. Equipment and schematic for measuring corona level.

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4.5.20 RF high potential withstanding voltage (see 3.23). The adapter shall be mated to a cabled connector in accordance with MIL-PRF-39012 (the cable shall be approximately 2 inches long). This assembly shall then be inserted into the high impedance circuit as shown on figure 6, or equivalent, and instantaneously subjected to the RF voltage and frequency specified (see 3.1) between the center contact and body of the adapters. The duration of the test shall be 1 minute. The RF voltage source shall be frequency stabilized and shall have an approximate pure sine wave output with minimum harmonic content. Means shall be provided to indicate disruptive discharge.

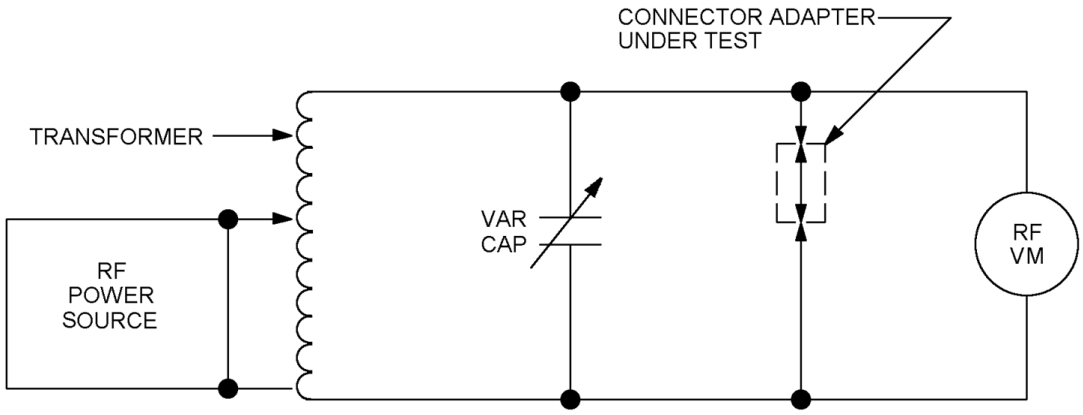


FIGURE 6. Circuit diagram for RF high potential withstanding voltage.

4.5.21 Corrosion (see 3.24). Adapters shall be tested in accordance with MIL-STD-202-101. The following details and exceptions shall apply:

a. Test condition letter: B.
b. Salt solution: 5 percent.

After exposure, adapters shall be washed, shaken and lightly brushed as specified in MIL-STD-202-101 and then permitted to dry for 24 hours at 40°C. Adapters shall then be examined for evidence of corrosion, pitting, and ease of coupling.

4.5.22 Coupling mechanism retention forces (see 3.25). The adapter body and coupling mechanism shall be respectively secured to the lower and upper jaws of a tensile tester in an appropriate manner. A tensile load shall be applied at a rate of approximately 100 pounds/minute up to the force as specified and held at that value for one minute (see 3.1). During the one minute of steadily applied force, the coupling mechanism shall be rotated with respect to the adapter body, two full revolutions in each direction.

4.5.23 Safety wire hole pullout (see 3.28). A single strand of safety wire shall be looped through the safety wire hole and secured to itself. Forces of 15 pounds (67 Newtons) minimum shall be applied to the safety wire pulling away from the adapter. One pull shall be parallel to the adapter axis and one pull perpendicular to the adapter axis (see figure 7). The safety wire shall be corrosion resistant steel .020 inch diameter (24 gauge) or .015 inch diameter, (27 gauge). This test is to be conducted under static conditions. All holes are to be tested individually.

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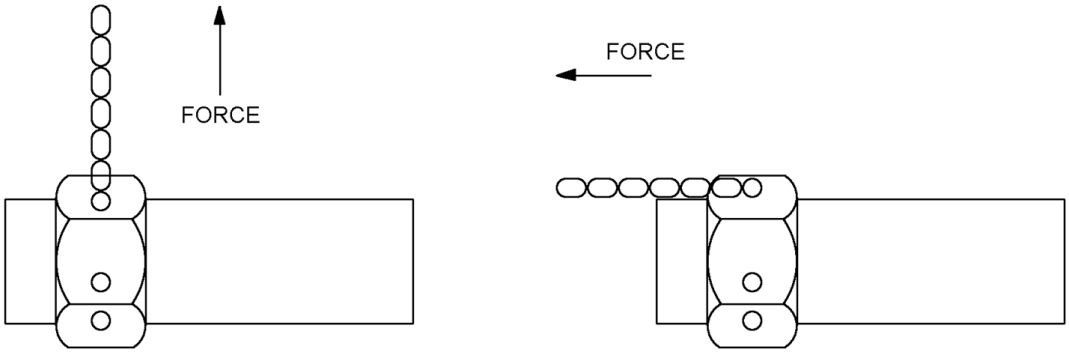


FIGURE 7. Safety wire hole pullout procedure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Connectors and fittings covered by this specification are intended for use in radio frequency application up to the frequency specified (see 3.1).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Title, number and date of the applicable detail specification.

c. Packaging requirements (see 5.1).

d. The complete PIN of the adapter or fitting ordered.

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the applicable Qualified Products List QPL No. 55339 whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus (DSCC-VQ), P.O. Box 3990, Columbus, Ohio 43218-3990.

6.4 Amendment notations. The margins of this amendment are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.5 Subject term (key word) listing:
Copper beryllium
Silver

6.6 Major quality issues.

a. An audit of the manufacturing plant indicates severe non-conformances to the QML/QPL/QTSL/QSLD program and/or applicable specifications. This includes repeat findings (non-conformances) found during periodic audits, which indicates that no corrective actions were implemented to clear the finding from previous audit.

b. The manufacturer's name appears on the "Consolidated List of Debarred, Ineligible, and Suspended Contractors".
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c. The manufacturer's QML/QPL/QTSL/QSLD product does not meet the quality, reliability, or performance requirements of the product or commodity specification and the manufacturer is unable to implement a suitable corrective action plan to return the product to compliance.

e. One or more conditions, under which qualification was granted, have been violated.

f. The manufacturer failed to notify and receive approval from the Qualifying Activity of a change in product design, material, product baselines, manufacturing locations, certified procedures and processes. It is still a violation of the program if a manufacturer notifies the Qualifying Activity after any of the aforementioned actions have been taken, and thus has failed to follow through with any necessary requalification of product (as determined by the Qualifying Activity).

g. A manufacturer fails to perform the required testing as determined by the applicable specifications and applicable program. This includes all Qualification, Conformance, and Periodic testing (Requalification/retention testing). A manufacturer must have objective evidence to support the completion of all required testing.

h. Manufacturer has been place on a Stop-Shipment status or in a Probationary status and has not provided the required information in the agreed-upon timeframes to become compliant to the applicable specification(s) and applicable government program.

i. A manufacturer has been in Stop-Shipment status for more than six months. Or a company has been placed on a Stop-Shipment one or more times within a 24-month period.

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j. A manufacturer incurs PQDRs or quality failures in the field for products manufactured under one of the subject government programs and are judged to be of a technically serious nature by the Qualifying Activity.

k. A manufacturer is under investigation or legal action is being taken by another government agency against the manufacturer for business practices which may have an effect on the integrity of the product produced under the applicable government program. (Example of Government agencies: DCMA, DOJ, IRS, BATFE)

l. A manufacturer fails to have in place, an effective quality system that can clearly demonstrate how the requirements of military specifications and elements of the applicable program are being met. The manufacturer fails to show controlled, repeatable operations and documented process flow for manufacturing products for one of the subject government programs. Critical elements of the applicable program, and its governing documents SD-6, and 4120.24 are being violated.

NOTE: Applicability of Criteria. Criteria applicable to placing a company on probation status and or a stop-ship, is also applicable to remove a company from the QML/QPL/QTSL/QSLD program or denying a company qualification to the program.

TABLE VIII. Cross-reference information

Superseding PIN M55339/	Superseded PIN or type designation	Superseding PIN M55339/	Superseded PIN or type designation
03-00027	MS90156 UG-27D/U	12-01138	MS90554 MS90266
04-00030	MS90159 UG-30 E/U		REB49272 UG-701A/U
05-00057	MS15507 UG-57 B/U		UG-1138A/U
06-00028	MS15506 REB49077 UG-28A/U	13-00492	MS35177 REB49098 UG-492D/U
06-00001	REB49080 UG-107B/U	14-00306	MS35368 UG-306B/U
07-00029	MS90576 RE49D258E UG-29C/U	15-00491	MS35176 UG-491B/U
08-00643	MS35288 UG-643/U	16-00914	MS35184 UG-914/U
09-00642	MS35287 UG-642A/U	17-00274	MS35173 UG-274C/U
10-00567	MS35322 UG-567A/U	18-01034 19-00565	UG-1034/U UG-565 A/U MS35324 REB49250
11-00566	MS35320 UG-566A/U	20-00201	UG-201A/U REB49324

TABLE VIII. Cross-reference information - Continued

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Superseding PIN M55339/	Superseded PIN or type designation
21-00564	MS35321 REB49247 UG-564A/U
22-00636	MS35283 REB49238 UG-636A/L
23-00635	MS35331 REB49237 UG-635/U
24-00335	REB49103 UG-335A/U
25-00606	51C12929 UG-606/U
28-30001	2084-8001-90 2084-8001-92 SF2991-6020
48-00002	M55339/48-00001
49-00349	UG-349/U

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
NASA – NA
DLA - CC

Preparing activity:
DLA – CC

(Project 5935-2023-001)

Review activities:
Army – AR, AT
Navy – AS, MC, OS
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil .

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